

Heuristic 1: Defined Data Scan

Scans Ghidra's typed data items once per binary. `s_*` labels yield strings directly; `PTR_s_*` labels are pointer-dereferenced.

```
From tailscale/tailscale — client/systray.setAppIcon (stripped build):
// Strings:
//  (@\textcolor{phase1col}{\textbf{s\_loading\_00f7a900}}@) = "loading"

C pseudocode — recovered string used in a comparison:
if ((in_stack_00000008 == s_loading_00f7a900)
    && (DAT_00f7a908 == cStack00000000000000010)) {
    tailscale_com_client_systray_startLoadingAnimation();
}

Ghidra typed s_loading as string data at 0xf7a900; the pipeline reads "loading" directly from that address.
```

Heuristic 2: Undefined Data Resolution

Resolves `PTR_s_*` / `s_*` references in C pseudocode absent from the data listing. Parses hex suffix from name; reads Go (`ptr`, `len`) pairs for precise, length-delimited extraction.

```
From usememos/memos — plugin/filter.NewAttachmentSchema (default build):
// Strings:
//  (@\textcolor{phase2col}{\textbf{PTR\_s\_attachmentindex\_html\_..\_01aad170}}@) = ["attachment", "memo_id"]
//  (@\textcolor{phase2col}{\textbf{PTR\_s\_attachmentindex\_html\_..\_01aacfc0}}@) = ["attachment", "filename",
//    "mime_type", "scalar", "string", "attachment", "type",
//    "create_time", "scalar", "timestamp", ...]

C pseudocode referencing the symbols (initializing map entries):
local_68 = PTR_s_attachmentindex_html_..\_01aacfc0;
uStack_60 = _UNK_01aacfc8;
...
runtime_mapassign(in_RDI, in_RSI, key, &PTR_DAT_01a76720);

Hex suffix _01aacfc0 parsed → memory at 0x1aacfc0 read as Go (ptr, len) pairs → 17 domain-specific strings recovered.
```

Heuristic 3: DAT Symbols

Resolves `DAT_*` labels (unclassified data). Detects paired string lengths from adjacent stack locals or function arguments.

```
From ollama/ollama — main.main.func1 in pull-progress (default build):
// Strings:
//  (@\textcolor{phase3col}{\textbf{DAT\_007ac68b}}@) = "Progress: status=%v, total=%v, completed=%v"

C pseudocode — used as a format string argument:
w.tab = (internal_abi_ITab *)0x2c; // length = 44
format.len = (int)&DAT_007ac68b;
format.str = extraout_RDX;
fmt_Fprintf(w, format, in_stack, 3, ...);

Hex address parsed from label. Length 0x2c (44) detected from adjacent assignment, matching the 44-byte format string.
```

Phase 4: Hex Literals

Resolves bare hex constants in C pseudocode that fall within `.rodata` address ranges.

```
From argoproj/argo-workflows — v1alpha1.ClusterWorkflowTemplate.GetResourceScope (stripped build):
// Strings:
//  (@\textcolor{phase4col}{\textbf{0x3171032}}@) = "cluster"

C pseudocode — bare hex address returned as a string pointer:
undefined8 ...GetResourceScope(void) {
    return 0x3171032;
}

Address 0x3171032 falls within .rodata. The decompiler shows only a numeric constant; Phase 4 resolves it to "cluster".
```